Oxford College of Emory University

Chapter 14, Part 1

Money, Liquidity, and Banking

Guilherme Paiva Pinto

October 27, 2026

France: Debt and Bank Liquidity

- **October 2026**: budget disputes and political uncertainty put pressure on French public finances.

Source: Agence France Tresor, daily TEC 10, October 22 and 27, 2026.
Source: Le Monde, French fiscal and political context, October 26, 2026.

France: Debt and Bank Liquidity

- **October 2026**: budget disputes and political uncertainty put pressure on French public finances.
- Ten-year borrowing indicator: **5.13% on October 22**; **4.94% on October 27**.

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France: Debt and Bank Liquidity

- **October 2026**: budget disputes and political uncertainty put pressure on French public finances.
- Ten-year borrowing indicator: **5.13% on October 22**; **4.94% on October 27**.
- Higher yields lower the market value of existing fixed-rate bonds. A bank selling them to meet withdrawals can realize losses.

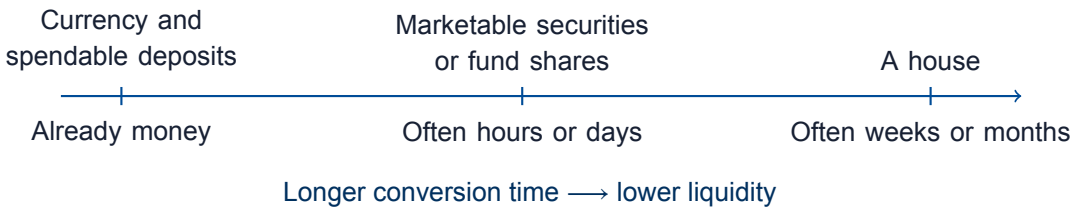
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Roadmap

1. Money, liquidity, and the money supply
2. Banks, reserves, and T-accounts
3. Bank runs and safeguards
4. Moral hazard and shadow banking

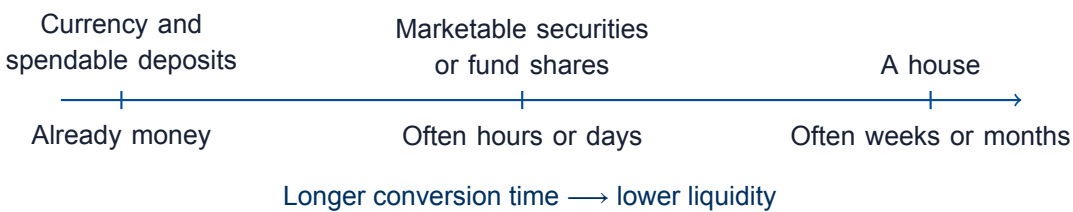
Money and Liquidity

- **Money**: assets that can readily be used to purchase goods and services.



Money and Liquidity

- **Money**: assets that can readily be used to purchase goods and services.
- **Liquidity**: how quickly an asset can become spendable money, with little cost or loss of value.



The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.

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The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

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The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.

The Functions of Money

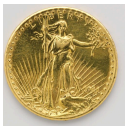
- **Medium of exchange**: an accepted way to pay for goods and services.
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- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.

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  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

Types of Money

What gives each item its value as money?



Gold coin, 1911

Commodity money
The item has value in other uses: its gold can be used outside payments.



Gold certificate, 1928



Modern U.S. dime

Gold coin: The Met, 1979.486.8, CC0. Certificate: U.S. Treasury, public domain. Dime: United States Mint image.

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A paper claim with a promise of redemption for a specified commodity.

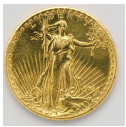


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Modern U.S. dime

Fiat money

Officially accepted money with no fixed promise of redemption for gold or silver.

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What Banks Do

- Banks **accept deposits** and provide payment services.

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- They keep part of their funding in **reserves** and lend a large portion or invest it in other assets.

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What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or invest it in other assets.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

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Bank Reserves

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- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

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T-Accounts

A **T-account** lists **assets on the left** and **liabilities on the right** to summarize a bank or business.

Assets		Liabilities	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Assets: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' equity.

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

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Reserves	$100,000		

Assets: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' equity.

Reserve ratio $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

Bank Runs

- A **bank run** occurs when many depositors withdraw because they fear the bank may fail.

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Bank Runs

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- The bank holds only part of its funding as immediately available reserves; loans cannot all be collected at once.

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Bank Runs

- A **bank run** occurs when many depositors withdraw because they fear the bank may fail.
- The bank holds only part of its funding as immediately available reserves; loans cannot all be collected at once.
- Forced asset sales can cause losses, increasing fear and triggering more withdrawals.

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Silicon Valley Bank: March 2023

- **2022**: higher rates reduced long-term securities' values; technology-sector customers drew down deposits.

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- **March 8–9, 2023**: SVB announced securities losses and a capital plan; the next day, deposits fell by more than $40 billion.

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- **March 10, 2023**: California closed the bank. Many large, uninsured and connected depositors made the run especially fast.

Source: Federal Reserve Board, SVB review, April 2023.

Bank Regulation: Deposit Insurance

- **Great Depression response**: the FDIC was created in **1933**; federal deposit coverage began in **1934**.

Source: FDIC, Deposit Insurance at a Glance.
Source: FDIC, historical timeline, 1933–1934.

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- Eligible deposits are insured up to **$250,000 per depositor, per insured bank, for each ownership category**.

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Bank Regulation: Deposit Insurance

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- Eligible deposits are insured up to **$250,000 per depositor, per insured bank, for each ownership category**.
- Insurance reduces the incentive to rush for the exit because a bank might fail.

Source: FDIC, Deposit Insurance at a Glance.
Source: FDIC, historical timeline, 1933–1934.

Bank Regulation: The Discount Window

- The Federal Reserve can lend to eligible banks against **pledged collateral**.

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Bank Regulation: The Discount Window

- The Federal Reserve can lend to eligible banks against **pledged collateral**.
- Temporary liquidity can help meet withdrawals and avoid hurried asset sales.
- The bank owes repayment; borrowing does not remove losses on its assets.

Source: Federal Reserve Board, Discount Window.

Interest on Reserve Balances

- **IORB**: interest the Fed pays on eligible reserve-account balances held at the Fed.

Source: Federal Reserve Board, Interest on Reserve Balances FAQ.

Interest on Reserve Balances

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Interest on Reserve Balances

- **IORB**: interest the Fed pays on eligible reserve-account balances held at the Fed.
- Reserves earn a return; IORB affects the incentive to hold them rather than acquire other assets.
- Changing IORB helps guide short-term market interest rates.

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Bank Regulation: Capital Requirements

- **Bank capital (equity)** is the value of assets minus liabilities.

Source: Federal Reserve Board, regulatory capital.

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- **Bank capital (equity)** is the value of assets minus liabilities.
- Regulators require a minimum capital cushion relative to a bank's exposures.
- Capital absorbs losses before creditors and gives owners more of their own funds to lose.

Source: Federal Reserve Board, regulatory capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.

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Moral Hazard

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- Deposit insurance can weaken depositors' incentive to monitor a bank. Owners may gain from risky bets while others bear part of the losses.

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Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.
- Deposit insurance can weaken depositors' incentive to monitor a bank. Owners may gain from risky bets while others bear part of the losses.
- **Capital requirements** put more of owners' funds at risk; supervision complements that incentive.

Source: Federal Reserve Board, regulatory capital.

Shadow Banking

- **Shadow banking**: credit intermediation outside ordinary deposit banking.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

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- Examples: **investment banks**, **money market funds**, hedge or credit funds, finance companies, and securitization vehicles.

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- Examples: **investment banks**, **money market funds**, hedge or credit funds, finance companies, and securitization vehicles.
- Some insurance-group activities also belong to this chain. Short-term funding of illiquid assets can create run-like risks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Shadow Banking in the Great Recession

Date	Institution	What happened
March 2008	Bear Stearns Investment bank	Funding run; acquired with Fed support.
September 15, 2008	Lehman Brothers Investment bank	Filed for bankruptcy.
September 16, 2008	AIG Insurance group	Collateral pressure; emergency Fed support.
September 16, 2008	Reserve Primary Fund Money market fund	Lehman losses; share value fell below $1.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

Source: Federal Reserve Board, AIG assistance, September 16, 2008.

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

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Reserves +$100; deposits +$100.

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What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Money and Bank Claims

- A deposit is money to its holder and a liability of the bank.

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- Part 2: the Fed, loan creation, settlement, repayment, and deposit expansion.

Oxford College of Emory University

Chapter 14, Part 2

Money Creation and the Federal Reserve

Guilherme Paiva Pinto

October 29, 2026

Roadmap

1. The Federal Reserve System
2. Loan creation, settlement, and repayment
3. A conditional deposit-expansion model
4. Practice and an individual checkpoint

The Federal Reserve System

Institution	Role
Board of Governors	Federal oversight and system responsibilities
12 Reserve Banks	Regional operations, financial services, and supervision
Federal Open Market Committee (FOMC)	Decisions about the monetary-policy stance

1913: the Federal Reserve Act established the Fed to address banking-system stresses.

Source: Federal Reserve Board, The Fed Explained.

Bank Accounting

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

Reserves and loans are assets. Deposits are liabilities. Equity is the residual.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank credits a new $120 loan to a customer's deposit.

Loan Origination

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Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

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The bank credits a new $120 loan to a customer's deposit.

Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

Reserves remain 200; equity remains 100.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
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Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Bank A: reserves −120; deposits −120.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
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The borrower pays $120 to a seller at Bank B.

Bank A: reserves −120; deposits −120.

Bank B: reserves +120; seller’s deposits +120.

The Banking System

Change since before the loan	Bank A	Bank B	Combined
Reserves	−120	+120	0
Loans	+120	0	+120
Deposits	0	+120	+120

The payment moves deposits; the loan created the additional $120 of deposits.

Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.

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Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.
- Public currency rises by $50, offsetting the fall in deposits.
- Currency + deposits is unchanged; the bank has less liquidity.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.

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- Loans fall by $50 and deposits fall by $50.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.
- Loans fall by $50 and deposits fall by $50.
- Currency + deposits falls by $50; reserves and equity do not change.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank writes off $80 of unrecoverable loans.

Loan Losses and Bank Capital

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Loans fall to 820; equity falls to 20.

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The bank writes off $80 of unrecoverable loans.

Loans fall to 820; equity falls to 20.

Deposits and reserves remain unchanged at the write-off.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.

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Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.
- **Liquidity and funding**: payments can leave the bank.
- **Risk and profitability**: borrowers must qualify, and lending must cover costs.

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A Reserve-Ratio Model

- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.

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- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.
- All loan proceeds are redeposited; no extra reserves or currency leakage.
- Banks lend each available amount; borrowers want the loans; other constraints do not bind.

Successive Deposit Rounds

Round	New deposit ($)	Reserves retained	New loan
1	100.00	10.00	90.00
2	90.00	9.00	81.00
3	81.00	8.10	72.90

The central bank first buys a $100 security from a nonbank seller, adding reserves and a deposit.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.

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- **Additional loans**: \$900. Total extra reserves: \$100.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.
- **Additional loans**: \$900. Total extra reserves: \$100.
- These are limiting model results under the stated conditions.

Depositing Existing Currency

A customer instead deposits $100 of currency already held outside banks.

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Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

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Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

With the same expansion conditions, deposits rise by 1,000 in total, but currency + deposits rises by 900.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.

Source: Federal Reserve Board, Reserve Requirements.

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- Banks still need liquidity and capital and must manage funding and credit risk.

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The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.
- Banks still need liquidity and capital and must manage funding and credit risk.
- $1/r$ at $r = 0$ does not predict infinite lending or an infinite money supply.

Source: Federal Reserve Board, Reserve Requirements.

Central-Bank Security Purchases

Transaction	Reserves	Nonbank deposits
Buy from a bank	Bank reserves rise	No immediate required change
Buy from a nonbank seller	Seller’s bank reserves rise	Seller’s deposit rises

Both transactions exchange assets; the immediate effect depends on the seller.

Balance-Sheet Practice

Use the worksheet's bank: reserves 150, loans 750, securities 100, deposits 900, equity 100.

Record a new $90 loan, then its payment to another bank.

For each event, verify assets = liabilities + equity and state what happens to total deposits.

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

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Loans −30 and equity −30.

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Loans −30 and equity −30.

Deposits and reserves do not change solely because of the write-off.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.

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- The multiplier depends on its model conditions.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.
- The multiplier depends on its model conditions.
- Chapter 15 connects central-bank policy, interest rates, spending, and AD.

Reference Definitions and Examples

Money, income, wealth, and bank accounting.

Money, Income, and Net Wealth

Concept	Meaning	Example
Money	Readily spendable assets	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 − $2,000 = $3,000

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank pay now?	Assets cannot all be sold quickly at little cost.
Solvency	Do assets cover liabilities?	Losses can exhaust equity.

Forced sales can turn liquidity pressure into losses.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

Reserves and deposits each rise by $200.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

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- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.

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A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.

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Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.